WORKED SOLUTION

STEP I 2019, Question 5

Pure • Vectors in Three Dimensions

STEP I 2019, Question 5

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Two vectors are equal exactly when their components agree, so every geometric condition in this question turns into a few simple equations in p, q, r and s. Coplanarity of four points is a scalar triple product (a 3×3 determinant) equal to zero. Once the conditions are written in coordinates, each part is short algebra.

Solution

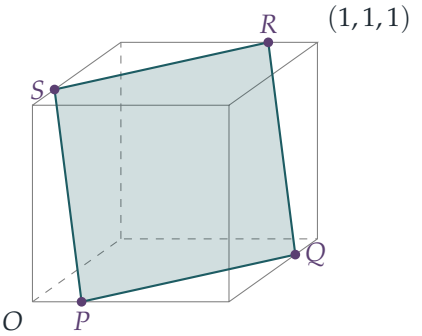
Part (i)

If $\overrightarrow{PQ} = \overrightarrow{SR}$, the opposite sides PQ and SR are parallel and of equal length, so $PQRS$ is a **parallelogram**. (Then $\overrightarrow{PS} = \overrightarrow{PQ} + \overrightarrow{QR} - \overrightarrow{SR}$ gives $\overrightarrow{PS} = \overrightarrow{QR}$ too.)

If also $|\overrightarrow{PQ}| = |\overrightarrow{PS}|$, two adjacent sides of the parallelogram are equal, and hence all four sides are equal: $PQRS$ is a **rhombus**.

Part (ii): the coplanarity condition

The points lie on edges of the unit cube, so $p, q, r, s \in [0, 1]$. Here P and Q lie in the face $z = 0$, and R and S in the face $z = 1$.



The four points are coplanar exactly when $\overrightarrow{PQ}$, $\overrightarrow{PS}$ and $\overrightarrow{PR}$ are linearly dependent, that is, when their scalar triple product is zero. With

$$\overrightarrow{PQ} = \begin{pmatrix} 1-p \\ q \\ 0 \end{pmatrix}, \qquad \overrightarrow{PS} = \begin{pmatrix} -p \\ s \\ 1 \end{pmatrix}, \qquad \overrightarrow{PR} = \begin{pmatrix} r-p \\ 1 \\ 1 \end{pmatrix},$$

expanding the determinant along the first row gives

$$\begin{vmatrix} 1-p & q & 0 \\ -p & s & 1 \\ r-p & 1 & 1 \end{vmatrix} = (1-p)(s-1) - q(-p-(r-p)) + 0$$
$$= (1-p)(s-1) + qr.$$

Setting this to zero gives $rq = (1-s)(1-p)$, as required.

If you prefer to avoid determinants, write the plane through P, Q, R as $\overrightarrow{OP} + \lambda\overrightarrow{PQ} + \mu\overrightarrow{PR}$, whose points are

$$(p + \lambda(1-p) + \mu(r-p),\ \lambda q + \mu,\ \mu).$$

For $S = (0, s, 1)$ to be on it we need $\mu = 1$ from the z-component, then $\lambda q = s - 1$ from the y-component, and $r + \lambda(1-p) = 0$ from the x-component. Multiplying the last equation by q and substituting gives $rq + (s-1)(1-p) = 0$, the same condition. (This uses P, Q, R not being collinear, which is true since R is in the top face and P, Q in the bottom.)

Part (ii)(a)

We have

$$\overrightarrow{PQ} = \begin{pmatrix} 1-p \\ q \\ 0 \end{pmatrix}, \qquad \overrightarrow{SR} = \begin{pmatrix} r \\ 1-s \\ 0 \end{pmatrix}.$$

The z-components already agree, so

$$\overrightarrow{PQ} = \overrightarrow{SR} \iff r = 1-p \text{ and } q = 1-s \iff p + r = 1 \text{ and } q + s = 1.$$

The centroid is

$$\tfrac{1}{4}(\overrightarrow{OP}+\overrightarrow{OQ}+\overrightarrow{OR}+\overrightarrow{OS}) = \tfrac{1}{4}\begin{pmatrix} p+1+r \\ q+1+s \\ 2 \end{pmatrix},$$

and the centre of the cube is $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$. The third components always agree, so the centroid is at the centre if and only if $p+1+r=2$ and $q+1+s=2$, which are the same two conditions $p+r=1$, $q+s=1$. Hence the two statements are equivalent.

(Notice that when $r=1-p$ and $s=1-q$ the coplanarity condition $rq=(1-s)(1-p)$ becomes $(1-p)q=q(1-p)$, which holds automatically: no configuration is ruled out.)

Part (ii)(b)

From (a), $\overrightarrow{PQ}=\overrightarrow{SR}$ gives $r=1-p$ and $s=1-q$. For the second condition compare squared lengths, using $\overrightarrow{PS}=(-p,\ s,\ 1)$ with $s=1-q$:

$$\begin{aligned} |\overrightarrow{PQ}|^2 = |\overrightarrow{PS}|^2 &\iff (1-p)^2+q^2 = p^2+(1-q)^2+1 \\ &\iff 1-2p = 2-2q \\ &\iff q = p+\tfrac{1}{2}. \end{aligned}$$

So

$$q = p+\tfrac{1}{2}, \qquad r = 1-p, \qquad s = \tfrac{1}{2}-p.$$

Since $q\le 1$ and $s\ge 0$, the points lie on the edges only for $0\le p\le \frac{1}{2}$.

Ask yourself Which two vectors give the angle PQR?

The angle at Q is between $\overrightarrow{QP}$ and $\overrightarrow{QR}$ (both pointing *away* from Q). Using the values just found,

$$\overrightarrow{QP} = \begin{pmatrix} p-1 \\ -p-\frac{1}{2} \\ 0 \end{pmatrix}, \qquad \overrightarrow{QR} = \begin{pmatrix} r-1 \\ 1-q \\ 1 \end{pmatrix} = \begin{pmatrix} -p \\ \frac{1}{2}-p \\ 1 \end{pmatrix}.$$

Then

$$\begin{aligned} \overrightarrow{QP}\cdot\overrightarrow{QR} &= -p(p-1)-(p+\tfrac{1}{2})(\tfrac{1}{2}-p) = p-p^2-(\tfrac{1}{4}-p^2) = p-\tfrac{1}{4}, \\ |\overrightarrow{QP}|^2 &= (1-p)^2+(p+\tfrac{1}{2})^2 = 2p^2-p+\tfrac{5}{4}, \\ |\overrightarrow{QR}|^2 &= p^2+(\tfrac{1}{2}-p)^2+1 = 2p^2-p+\tfrac{5}{4}. \end{aligned}$$

The two lengths are equal, as they must be in a rhombus, so

$$\cos PQR = \frac{p-\frac{1}{4}}{2p^2-p+\frac{5}{4}} = \frac{4p-1}{5-4p+8p^2},$$

on multiplying top and bottom by 4. This is the required result.

The square. A rhombus is a square exactly when its angles are right angles, i.e. $\cos PQR = 0$. The denominator $8p^2 - 4p + 5 = 8(p - \frac{1}{4})^2 + \frac{9}{2}$ is always positive, so this happens only for $4p - 1 = 0$:

$$p = \tfrac{1}{4}, \qquad q = \tfrac{3}{4}, \qquad r = \tfrac{3}{4}, \qquad s = \tfrac{1}{4}.$$

The side length squared is then

$$2 \cdot \tfrac{1}{16} - \tfrac{1}{4} + \tfrac{5}{4} = \tfrac{9}{8},$$

so each side has length $\sqrt{9/8} = \frac{3\sqrt{2}}{4}$. To compare with $\frac{21}{20}$, compare squares of these positive numbers:

$$\left(\tfrac{21}{20}\right)^2 = \tfrac{441}{400} < \tfrac{450}{400} = \tfrac{9}{8}.$$

Hence the side of the square is greater than $\frac{21}{20}$. (In fact it is about 1.0607.)

Checks

- At $p = \frac{1}{4}$ the four points are $(\frac{1}{4}, 0, 0)$, $(1, \frac{3}{4}, 0)$, $(\frac{3}{4}, 1, 1)$, $(0, \frac{1}{4}, 1)$; their average is $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, as (a) predicts, and $\overrightarrow{QP} \cdot \overrightarrow{QR} = \frac{3}{16} - \frac{3}{16} = 0$ directly.
- At $p = 0$ the formula gives $\cos PQR = -\frac{1}{5}$. Directly, $\overrightarrow{QP} = (-1, -\frac{1}{2}, 0)$ and $\overrightarrow{QR} = (0, \frac{1}{2}, 1)$, with dot product $-\frac{1}{4}$ and squared lengths $\frac{5}{4}$, so $\cos PQR = -\frac{1}{5}$. ✓
- The coplanarity identity, the equivalence in (a) and the formula for $\cos PQR$ were all confirmed by computer with exact arithmetic on many random values.

Common slips

- Using $\overrightarrow{QP}$ and $\overrightarrow{RQ}$ (one vector pointing in, one out) for the angle at Q: this flips the sign of the cosine.
- Mixing up the order of letters: $\overrightarrow{SR}$ is $R - S$, not $S - R$.
- Claiming $\frac{3\sqrt{2}}{4} > \frac{21}{20}$ from a calculator value without an exact comparison; squaring both sides keeps it rigorous because both are positive.

Beyond STEP

The determinant test for coplanarity is the three-dimensional case of a general fact: vectors are linearly dependent exactly when the determinant of the matrix they form vanishes. The plane sections of a cube are a rich source of examples too (a plane through the centre perpendicular to a long diagonal cuts out a regular hexagon). Strang's *Introduction to Linear Algebra* develops determinants, volumes and linear dependence from this geometric point of view.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.